

SMART SAFETY COMPANION

1. Project Overview

Smart Safety Companion is an Android-based prototype application developed using MIT App Inventor with the objective of improving personal safety.

The application makes use of inbuilt smartphone sensors such as the accelerometer sensor, location sensor, and orientation sensor.

These sensors continuously monitor the movement, position, and direction of the mobile device.

Whenever abnormal or sudden movement is detected, the system assumes a possible unsafe situation and generates an emergency alert.

This project demonstrates how sensor-based mobile applications can be used for real-time safety monitoring.

2. Problem Identification

In many real-life situations such as accidents or sudden emergencies, a person may not be able to manually call for help.

Delay in communication and lack of accurate location information can increase the risk.

There is a need for a simple and automatic safety system that can detect abnormal situations and provide useful information such as location and direction.

This project aims to address this problem using mobile sensor technology.

3. Proposed Solution

The proposed solution is a mobile application that continuously observes motion data from the accelerometer sensor.

When the motion intensity exceeds a predefined limit, the application treats it as an abnormal condition.

At this point, the system automatically captures the user's location, device direction, and current time.

An emergency alert message is then generated and displayed.

A cooldown mechanism is used to prevent repeated alerts within a short period.

4. Development Environment

Software Platform:

MIT App Inventor is used to design the user interface and implement backend logic using block programming.

Hardware Platform:

An Android smartphone is required to test and run the application.

Sensors Used:

- Accelerometer Sensor for motion detection
- Location Sensor for GPS-based position tracking
- Orientation Sensor for identifying device direction

Components Used:

- Clock for time capture and delay control
- Buttons for user interaction
- Labels for displaying information
- TextBox for entering emergency contact number

5. Application Interface

The application interface is designed to be simple and user-friendly.

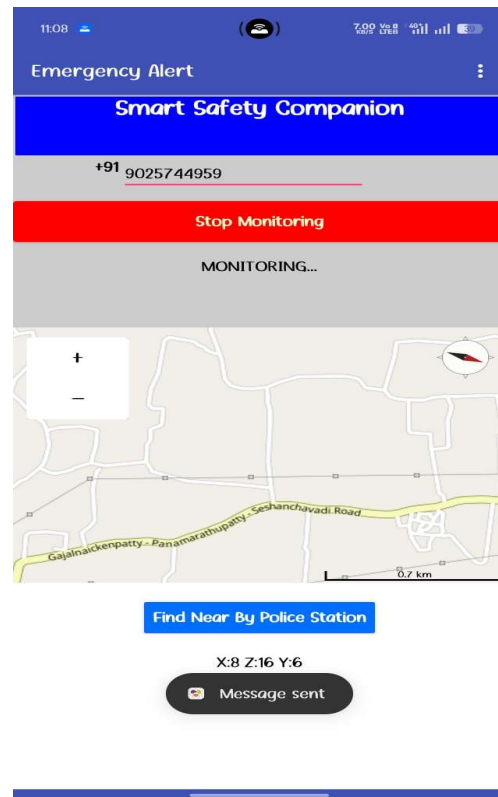
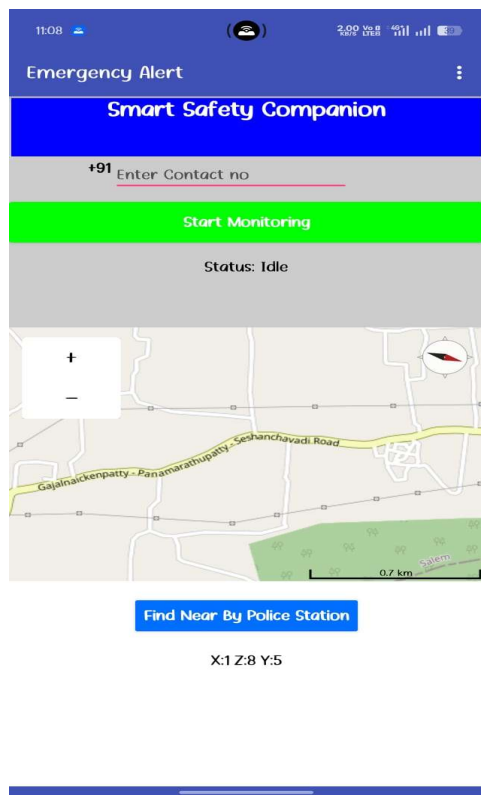
It includes a TextBox to enter the emergency contact number.

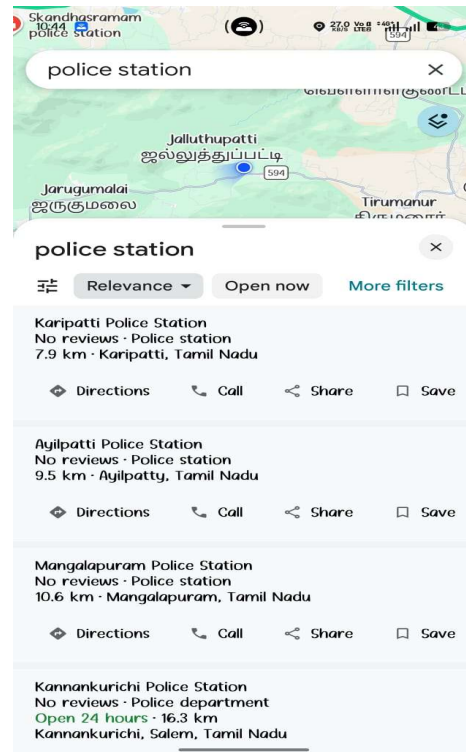
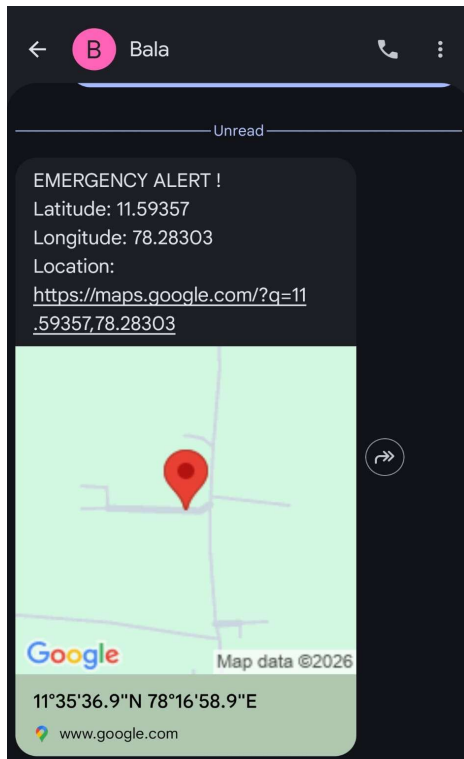
The Start Monitoring button is used to begin sensor monitoring.

The Stop Monitoring button is used to stop all sensor activities.

Several labels are provided to display accelerometer values, shake intensity, latitude, longitude, device direction, system status, and alert message.

The clear layout helps users understand the system status easily during emergencies.





6. Working Methodology

When the user presses the Start Monitoring button, all sensors are activated.

The accelerometer continuously provides X, Y, and Z axis values.

The shake intensity is calculated using the formula:

$$\text{Shake Value} = |X| + |Y| + |Z|$$

If the calculated shake value exceeds the preset threshold, the system detects an abnormal motion.

The application then captures the current location using the location sensor.

The orientation sensor is used to determine the device direction.

The current time is obtained using the Clock component.

An emergency alert message is generated and displayed.

A cooldown delay is applied to prevent multiple alerts.

Pressing the Stop Monitoring button disables all sensors.

7. Backend Implementation

The backend logic of the Smart Safety Companion is implemented using block-based programming in MIT App Inventor.

The backend is responsible for sensor control, data processing, emergency detection, and alert generation.

Start and Stop Control:

The Start button enables the accelerometer, location, and orientation sensors.

The Stop button disables all sensors and stops data processing.

Motion Detection:

Accelerometer values from all three axes are combined to calculate shake intensity.

This value is continuously compared with a threshold to detect abnormal motion.

Location and Direction Capture:

The location sensor provides real-time latitude and longitude values.

The orientation sensor provides the direction of the device.

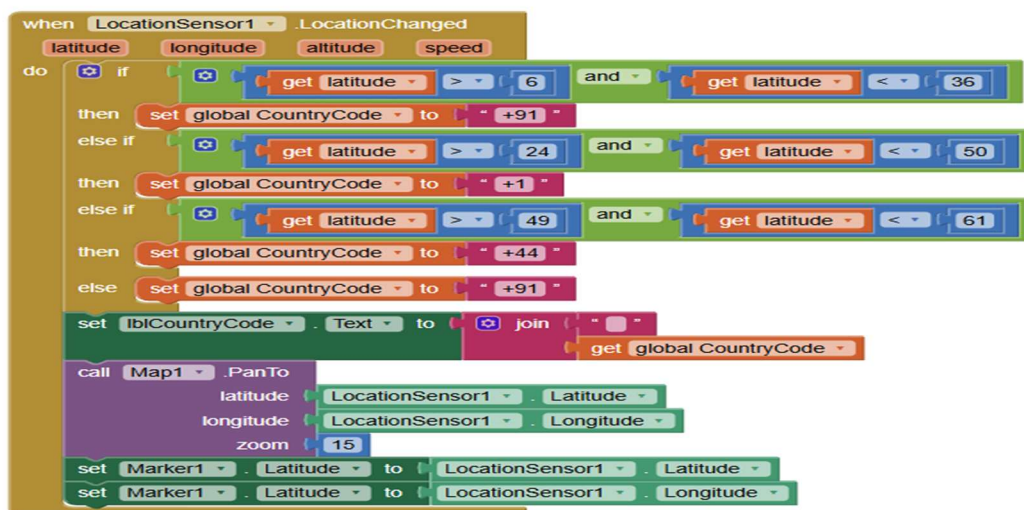
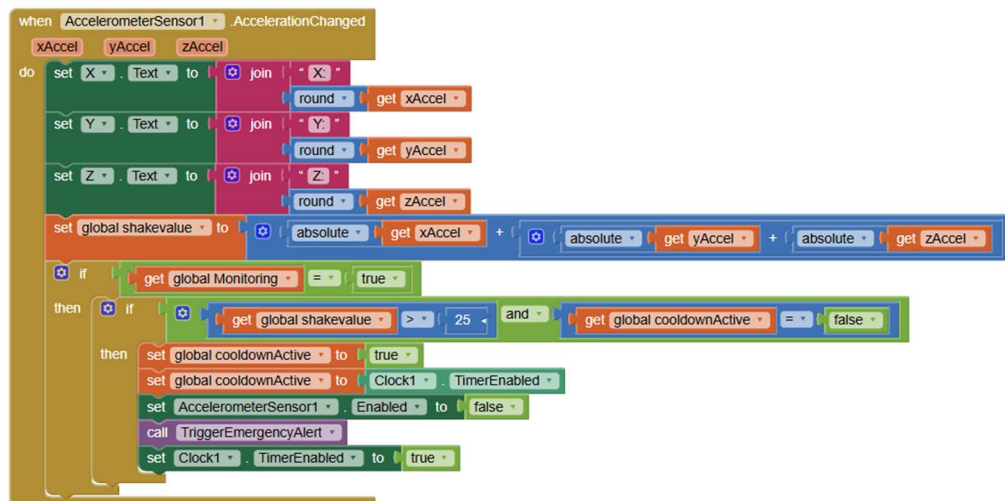
These values are displayed and used in the alert message.

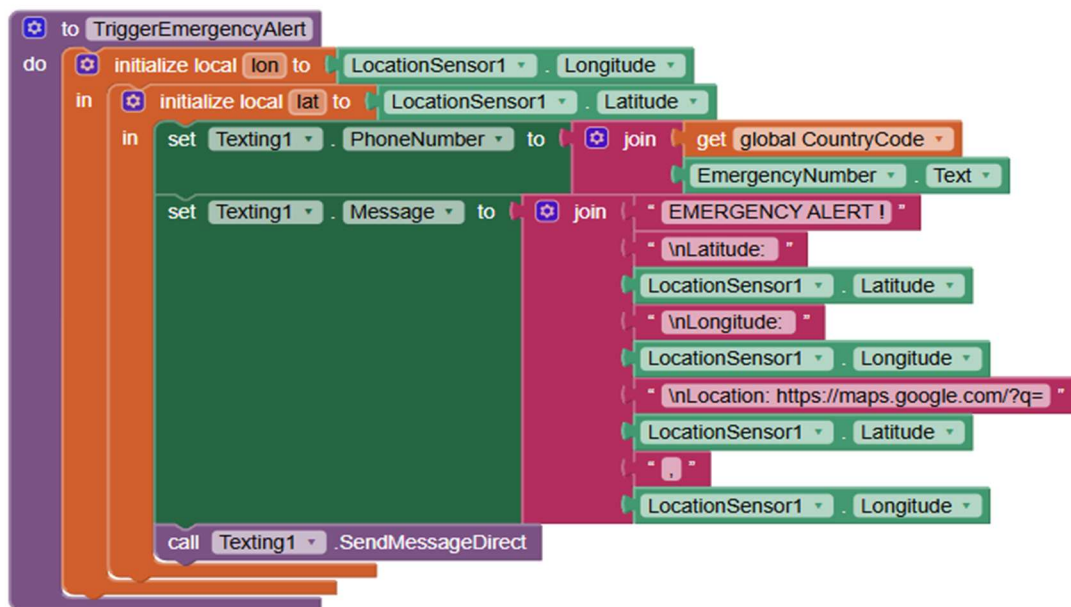
Alert Generation and Cooldown:

Once an emergency is detected, the alert message is created using collected data.

The Clock component manages the cooldown period to avoid repeated alerts.

(Screenshots of block implementation)





8. Advantages

- The system is easy to use and does not require complex operations
- It provides automatic detection of abnormal situations
- Real-time location tracking improves response efficiency
- The cooldown mechanism reduces false alerts
- The application is useful for personal safety purposes

9. Limitations

- The application works only on real Android devices
- Sensor and GPS permissions are required for proper functioning
- Accuracy depends on the quality of the mobile device sensors

10. Conclusion

The Smart Safety Companion project successfully demonstrates the use of smartphone sensors for safety applications.

The system effectively detects abnormal motion, captures essential information such as location and direction, and generates emergency alerts.

This project highlights the potential of MIT App Inventor in developing practical and real-world Android applications focused on personal safety.